

Exact entropy duality fails in dimension two

A finite-rank counterexample to arXiv:1703.01418

Automated mathematics discovery run `fa_banach_001`
Lane 2, candidate for human review

21 July 2026

Status: candidate counterexample, likely valid. Over real Banach spaces, the coordinate identity $T : \ell_1^2 \rightarrow \ell_2^2$ is finite rank and compact, but its second entropy number differs from that of its adjoint. We compute both values exactly by matching two-ball covers and five-point odd-cycle lower bounds.

1 Source question

Deepesh and Kiran Kumar ask whether

$$\varepsilon_k(T) = \varepsilon_k(T^*)$$

for finite-rank or compact operators between Banach spaces, and repeat the identity as Problem 4.1(1) for arbitrary Banach spaces [1]. The question appears in Remark 4.4 and Problem 4.1(1) on PDF page 9:

Remark 4.4. We would like to make the remark that if one requires to prove the equality $\varepsilon_k(T) = \varepsilon_k(T^*)$ for operators between certain special Banach spaces (like ℓ_p , $1 \leq p < \infty$, where one has finite rank projections of norm 1 which converges to identity operator in SOT/WOT), it is sufficient to prove the result for compact operators or for finite rank operators, in view of Theorem 2.3. However, it is still an open problem to prove the equality $\varepsilon_k(T) = \varepsilon_k(T^*)$ for finite rank/compact operators between Banach spaces.

4.1. Some of the important problems.

- (1) Consider the original problem by Carl, it is still an open problem to show the identity $\varepsilon_k(T) = \varepsilon_k(T^*)$ for bounded operators on arbitrary Banach spaces. Even it is not clear whether $\varepsilon_k(T^*) = O(n^{-\alpha})$ if $\varepsilon_k(T)$ is so in the general setting. However, since our techniques are applicable for reflexive Banach spaces with suitable projections, it suffices to show these for compact operators on the such spaces where we are able to approximate bounded operators by sequence of compact operators (in WOT).

We give a two-dimensional negative answer to the literal exact-identity question.

2 Definition and counterexample

For a bounded operator $S : X \rightarrow Y$, the source defines

$$\varepsilon_n(S) = \inf \left\{ r > 0 : S(B_X) \subseteq \bigcup_{j=1}^N (y_j + rB_Y) \text{ for some } N \leq n \right\}.$$

Thus $\varepsilon_2(S)$ is the optimal common radius of two balls covering $S(B_X)$.

Theorem 1 (Exact two-dimensional separation). *Let*

$$T : \ell_1^2(\mathbb{R}) \longrightarrow \ell_2^2(\mathbb{R}), \quad T(x_1, x_2) = (x_1, x_2).$$

Then T has rank two, T^ is the coordinate identity $\ell_2^2(\mathbb{R}) \rightarrow \ell_\infty^2(\mathbb{R})$, and*

$$\begin{aligned} \varepsilon_2(T) &= \sqrt{\frac{5}{8}}, \\ \varepsilon_2(T^*) &= \frac{1 + 1/\sqrt{2}}{2}. \end{aligned}$$

In particular,

$$\varepsilon_2(T) < \varepsilon_2(T^*),$$

so exact entropy-number duality fails even for a finite-rank compact operator between two-dimensional Banach spaces.

3 Proof intuition

The operator T asks for a two-disk cover of the planar diamond B_1^2 , whereas T^* asks for a two-axis-parallel-square cover of the Euclidean disk B_2^2 . Each body admits a simple optimal cover by two congruent balls.

Optimality is certified without optimization theory. On each boundary we select five points arranged cyclically so that consecutive points are at least one ball-diameter apart. If two smaller balls covered the five points, assigning each point to one of its covering balls would give a proper two-colouring of a five-cycle, which is impossible. The two geometries give different exact radii.

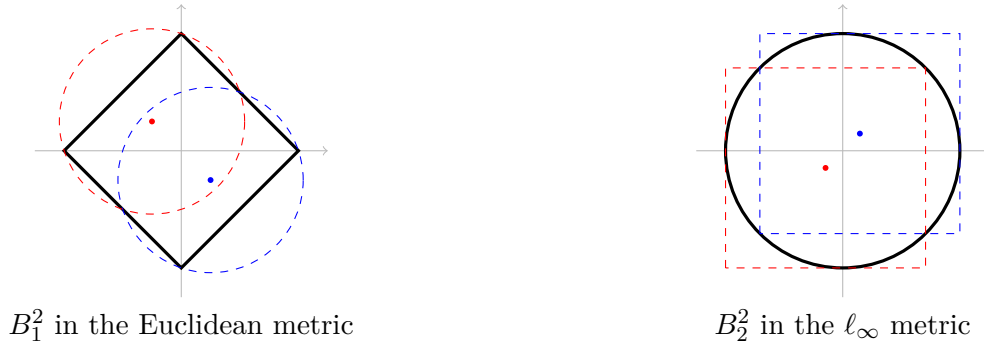


Figure 1: The explicit two-ball covers. Dashed circles and squares are target-norm balls.

4 The odd-cycle obstruction

Lemma 2. *Let K be a subset of a metric space. Suppose that $z_1, \dots, z_{2j+1} \in K$ and, cyclically,*

$$d(z_i, z_{i+1}) \geq 2r.$$

Then K cannot be covered by two balls of radius strictly less than r . Consequently its second entropy number is at least r .

Proof. If two balls of radius $r' < r$ covered the points, assign to every z_i one of the balls containing it. Two cyclic neighbours cannot receive the same ball because that ball has diameter at most $2r' < 2r$. This would properly two-colour an odd cycle, a contradiction. $\square$

5 Proof of Theorem 1

5.1 The diamond in Euclidean distance

Set

$$K = B_1^2 = \{(x, y) : |x| + |y| \leq 1\}, \quad r = \sqrt{\frac{5}{8}}, \quad c = \left(\frac{1}{4}, -\frac{1}{4}\right).$$

The half-diamond $K_+ = K \cap \{x \geq y\}$ is the convex hull of

$$(1, 0), \quad (0, -1), \quad \left(\frac{1}{2}, \frac{1}{2}\right), \quad \left(-\frac{1}{2}, -\frac{1}{2}\right).$$

Each of these four vertices is at Euclidean distance exactly r from c . Since a Euclidean ball is convex,

$$K_+ \subseteq c + rB_2^2.$$

The other half is $K_- = -K_+$, so it is contained in $-c + rB_2^2$. Hence $\varepsilon_2(T) \leq r$.

For the reverse bound, take the following five boundary points of K , in cyclic order:

$$(1, 0), \quad \left(-\frac{1}{2}, \frac{1}{2}\right), \quad (0, -1), \quad \left(\frac{1}{2}, \frac{1}{2}\right), \quad (-1, 0).$$

The first four successive Euclidean distances are

$$\sqrt{\frac{5}{2}} = 2r,$$

and the last-to-first distance is $2 \geq 2r$. Lemma 2 gives $\varepsilon_2(T) \geq r$. Therefore

$$\varepsilon_2(T) = \sqrt{\frac{5}{8}}.$$

5.2 The Euclidean disk in the supremum metric

Under the canonical finite-dimensional dualities,

$$T^* : \ell_2^2 \longrightarrow \ell_\infty^2$$

is again the coordinate identity. Put

$$a = \frac{1}{\sqrt{2}}, \quad s = \frac{1+a}{2}, \quad d = \left(\frac{1-a}{2}, \frac{1-a}{2}\right).$$

The two closed ℓ_∞ -balls of radius s centred at d and $-d$ are the squares

$$[-a, 1]^2 \quad \text{and} \quad [-1, a]^2.$$

Their union contains B_2^2 . Indeed, a point of $[-1, 1]^2$ outside both squares must have one coordinate strictly less than $-a$ and the other strictly greater than a . Its squared Euclidean norm is then strictly larger than $2a^2 = 1$. Thus $\varepsilon_2(T^*) \leq s$.

For the lower bound, use the five points of the Euclidean unit circle

$$(1, 0), \quad (-a, a), \quad (0, -1), \quad (a, a), \quad (-1, 0).$$

The first four successive ℓ_∞ -distances are $1 + a = 2s$, while the last-to-first distance is $2 \geq 2s$. By Lemma 2, $\varepsilon_2(T^*) \geq s$. Hence

$$\varepsilon_2(T^*) = \frac{1 + 1/\sqrt{2}}{2}.$$

Finally,

$$\left(\frac{1 + 1/\sqrt{2}}{2} \right)^2 - \frac{5}{8} = \frac{\sqrt{2} - 1}{4} > 0,$$

which proves the strict inequality and completes the counterexample.

6 Verification, scope, and novelty check

The accompanying script `code/verify_exact_values.py` checks every listed vertex and cycle distance, then samples 80,004 boundary points against the two explicit covers. It reproduces the two exact radii numerically. The computation is only a sanity check; the proof above is symbolic.

The counterexample is over real Banach spaces. This suffices to refute the source’s literal assertion for arbitrary Banach spaces. If “Banach spaces” was intended to mean complex spaces only, that narrower version is not settled here. The example also does not contradict the source’s Hilbert-to-Hilbert theorem: the domain ℓ_1^2 is not Hilbert. Finally, it does not address the classical *essential* or asymptotic duality problem for entropy numbers, nor the separate rate-transfer question in Problem 4.1(1); it only disproves pointwise exact equality.

The run’s registry, solution, attempt, and proof-gap indexes were searched by arXiv id, title, and the phrases “entropy numbers adjoint exact equality” and “finite rank entropy duality”. Bounded primary-source web searches used those phrases and the two formal identities $\ell_1^2 \rightarrow \ell_2^2$ and $\ell_2^2 \rightarrow \ell_\infty^2$. They found the source and classical literature on essential/asymptotic duality, but no occurrence of this exact two-dimensional counterexample. This is not an exhaustive novelty certification.

References

- [1] K. P. Deepesh and V. B. Kiran Kumar, *Approximation of Entropy Numbers*, arXiv:1703.01418 (2017).